

MACHINE LEARNING

LIFESTYLE PREDICTION CLASSIFIER

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Rationale: Obesity is a complex public health challenge driven by a combination of genetic predispositions, dietary habits, and physical conditions. While clinical risk factors are often assessed individually, understanding whether isolated indicators—such as family history—suffice for prediction compared to comprehensive behavioral and physiological profiles is critical for developing effective classification models.

Project's aim: This project investigates obesity prediction through two primary machine learning frameworks: first, evaluating whether a binary approach can accurately identify obese individuals and determining if family history alone is a sufficient attribute for prediction; second, assessing whether a multi-class approach can accurately categorize an individual's specific obesity level (NObeyesdad) based on eating habits and physical condition.

Approach: The methodology utilized both binary and multi-class classification workflows implemented in Python. The approach involved data preprocessing, feature selection using variance thresholds, and training supervised learning models—principally Random Forest classifiers, alongside comparisons with support vector machines and K-nearest neighbors. Model evaluation was conducted using performance metrics such as accuracy, classification reports, confusion matrices, and ROC-AUC curves, followed by systematic hyperparameter tuning via grid search to analyze model stability and boundary behavior.

Results: The binary classification model achieved an AUC of 0.97 using a multi-variable feature set, whereas a single-feature baseline relying strictly on family history yielded an AUC of 0.65, demonstrating that isolated genetic or environmental risk factors are insufficient on their own. For the multi-class framework, a Random Forest model achieved an overall accuracy of 0.83 (macro average: 0.84), with errors predominantly concentrated among adjacent, overlapping intermediate weight categories. Furthermore, hyperparameter tuning revealed that default parameters were already robustly optimized for the dataset's complex boundaries.

Resources: full pipeline and results can be found at

<https://github.com/alessiavisani/ML-LifestylePrediction-Classifier>

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Project's aim & project's research questions

This study aims to develop a robust machine learning framework capable of discriminating unseen individuals based on a curated set of attributes, including a family history of being overweight, water consumption frequency, physical activity levels, and primary means of transportation. By leveraging these behavioral and physiological features, the project seeks to establish a reliable classifier to assist in categorizing individuals into distinct weight classes, thereby providing deeper insights into broader population health trends.

- **BINARY CLASSIFICATION:** *Can we accurately predict whether someone falls into an 'Obese' category using a binary classification approach, and is family history alone sufficient to make that prediction?*
- **MULTI-CLASS CLASSIFICATION:** *Can we accurately predict an individual's obesity level category (NObesidad) based on their eating habits and physical condition?*

1. Importing the data & the libraries

The dataset was retrieved from UCI Machine Learning Repository(1) which is a collection of databases, domain theories, and data generators that are used by the machine learning community for the empirical analysis of machine learning algorithms. This dataset includes data for the estimation of obesity levels in individuals from the countries of Mexico, Peru and Colombia, based on their eating habits and physical condition. The data contains 17 attributes and 2111 records, the records are labeled with the class variable NOesity (Obesity Level), that allows classification of the data using the values of Insufficient Weight, Normal Weight, Overweight Level I, Overweight Level II, Obesity Type I, Obesity Type II and Obesity Type III. Further insights on the data can be found at <https://pmc.ncbi.nlm.nih.gov/articles/PMC6710633/> (2).

In order to perform the analysis as well as the visualization of the results different python libraries and Scikit-Learn modules were utilized, with this we mention:

- pandas
- numpy,
- %matplotlib inline
- matplotlib.pyplot
- seaborn
- sklearn.preprocessing : LabelEncoder
- sklearn.feature_selection : VarianceThreshold
- sklearn.model_selection : train_test_split, GridSearchCV
- sklearn.ensemble : RandomForestClassifier
- sklearn.metrics : classification_report, accuracy_score , log_loss, AUC, roc_curve, ConfusionMatrixDisplay, confusion_matrix
- plotly : plotly.graph_objects, plotly.express
- sklearn.svm : SVC
- sklearn.neighbors : KNeighborsClassifier

The dataset was downloaded in CSV format and hosted publicly on a personal GitHub repository. Following the initial importation of necessary Python libraries, the data was retrieved via its direct URL and structured into a pandas DataFrame for subsequent preprocessing and analysis. Correct loading of the data into the DataFrame was investigated with `data.info()` command ; The dataset consists of 2,111 entries and 17 columns, exhibiting no missing values. The features comprise a mix of numerical and object data types; since the data originates from a CSV format, the object types correspond directly to text attributes.

2. EDA - exploratory data analysis

In order to understand the behaviour and type of the data, exploratory data analysis practices were utilized as : *“The primary aim with exploratory analysis is to examine the data for distribution, outliers and anomalies to direct specific testing of your hypothesis. It also provides tools for hypothesis generation by visualizing and understanding the data usually through graphical representation. EDA aims to assist the natural patterns recognition of the analyst. (...) EDA is a fundamental early step after data collection and pre-processing , where the data is simply visualized, plotted, manipulated, without any assumptions, in order to help assessing the quality of the data and building models. ”*(3).

Firstly, I proceeded in inspecting the `value_counts()` for each entry in the columns, in particular the data was organized as:

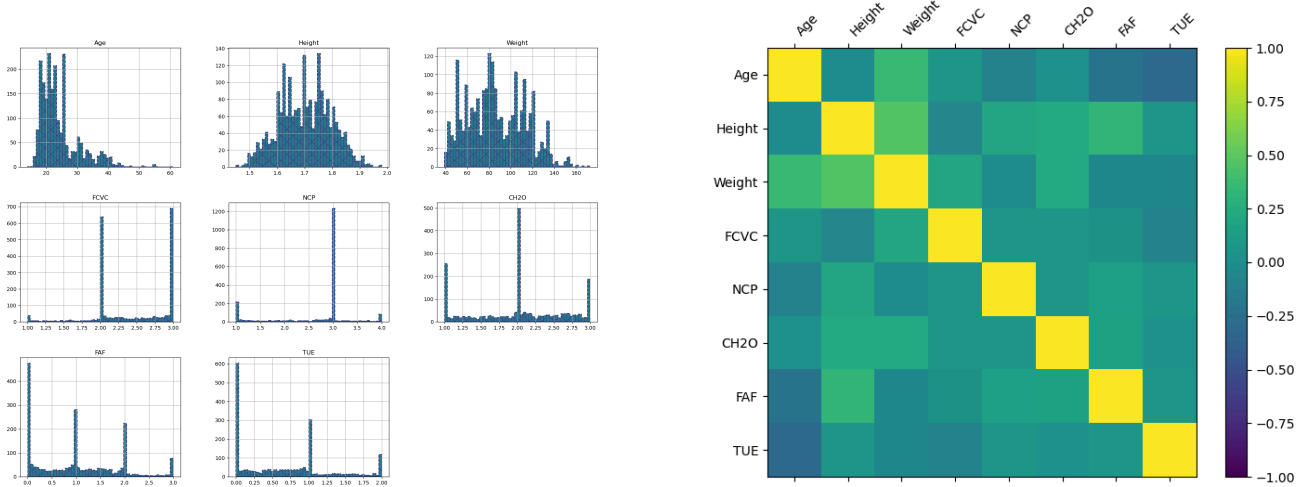
- gender (categorical): male,female
- age (continuous)
- height (continuous)
- weight(continuous)
- family history overweight (binary-boolean): yes, no
- FAVC- high caloric intake - (binary -boolean): yes, no
- FCVC - intake of vegetables - (binary-boolean):yes,no
- NCP - amount of meals within a day - (continuous)
- CAEC - consumption of food in between meals - (categorical)
- SMOKE (binary - boolean): yes, no
- CH20 - daily consumption of water- (continuous)
- SCC - monitoring calories- (binary boolean): yes, no
- FAF - frequency of physical activity- (continuous)
- TUE - time of use of technological devices - (integer)
- CALC - alcohol consumption- (categorical)
- MTRANS - means of transport - (categorical)
- NObeyesdad - obesity level > **TARGET FEATURE** - (categorical): Obesity_Type_I, Obesity_Type_III,Obesity_Type_II,Overweight_Level_I,Overweight_Level_II, Normal_Weight, Insufficient_Weight

Outcome of the investigation was that the target variable (`NObeyesdad`) demonstrates a well-balanced distribution across its seven distinct weight and obesity classes (ranging from 272 to 351 instances each), establishing an ideal foundation for predictive modeling. Furthermore, the demographic and lifestyle features highlight a balanced gender representation alongside behavioral trends within the population—such as a high prevalence of a family history of overweight status, frequent high-calorie food consumption (FAVC), and minimal tobacco use (SMOKE). Finally, transportation habits (MTRANS) show that public transit and automobiles dominate as the primary commuting methods, providing valuable context for subsequent feature engineering and exploratory data analysis.

Through usage of the command `data.describe()` it was possible to highlight mean and standard deviation for some classes such as age and weight/height noting that the features exhibit distinct scales and ranges—for instance, weight spans across a wide distribution reflecting the diverse body types and obesity categories present in the population, while variables like age capture a specific demographic range of the surveyed individuals.

Data was investigated for correlation and skewness using a correlation matrix approach; the decision to employ the Spearman correlation coefficient rather than Pearson is justified by the non-normal distribution and skewness of key numerical features, such as age and weight, where significant discrepancies between mean and median values are evident. While the Pearson coefficient is restricted to linear dependencies and is sensitive to outliers, Spearman evaluates monotonic relationships using ranked data. This approach is more robust against distributional skewness and effectively captures non-linear associations—such as the pronounced relationship between age and weight—that would otherwise be underestimated or overlooked by a linear metric.

Some visualization of the distribution of values within the classes using histograms as well as the contribution provided by the correlation matrix



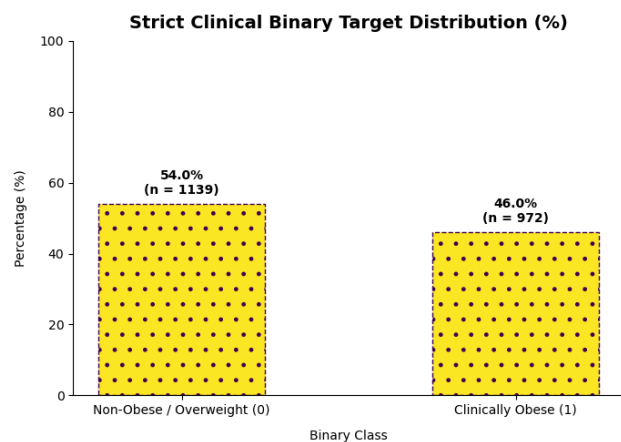
While the histograms on the left show as aforementioned the different distribution of such classes, the correlation matrix highlights the presence of variables correlated to one another. In our interest we see that Weight-Age and Weight-Height are tightly correlated therefore careful choices are to be made in order to account for the relatedness of data.

3. Analytical decisions

To properly address both research objectives, the workflow branches during the analytical decisions phase into two distinct dataset versions: one optimized for binary classification and another for multi-class classification. This approach allows the target variable (NObesyedad) to be explicitly transformed for each context—collapsing into a two-class label for the binary task while preserving core categories for the multi-class task. By maintaining separate data pipelines, we ensure that subsequent modeling and evaluation accurately reflect how tree-based algorithms handle varying degrees of class complexity without cross-contamination or data leakage.

3.1 creating the binary dataset

For the binary classification pipeline, the target variable NObesyedad is mapped using strict clinical guidelines. Individuals classified under Obesity Type I, Type II, and Type III are grouped into the positive class (1), representing clinically diagnosed obesity ($BMI \geq 30$). Conversely, individuals categorized as Underweight, Normal Weight, and Overweight are grouped into the negative class (0). This establishes a definitive medical boundary, allowing us to evaluate how effectively tree-based models can separate high-risk obese individuals from the rest of the population using behavioral and lifestyle features.



3.2 creating the multiclass dataset

Dropping height and weight from the multi-class feature set is a crucial analytical decision designed to prevent data leakage and preserve the integrity of the research question. Because the target variable (NObeyesdad) is mathematically derived directly from Body Mass Index (BMI)—which is calculated using an individual's height and weight—retaining these physical measurements would provide the model with an artificial shortcut. Instead of learning patterns from behavioral and lifestyle attributes, algorithms would instantly rely on these direct physical metrics to near-perfect accuracy. Removing height and weight ensures that the model is genuinely tested on its ability to predict distinct weight categories based solely on demographics, dietary habits, and lifestyle factors.

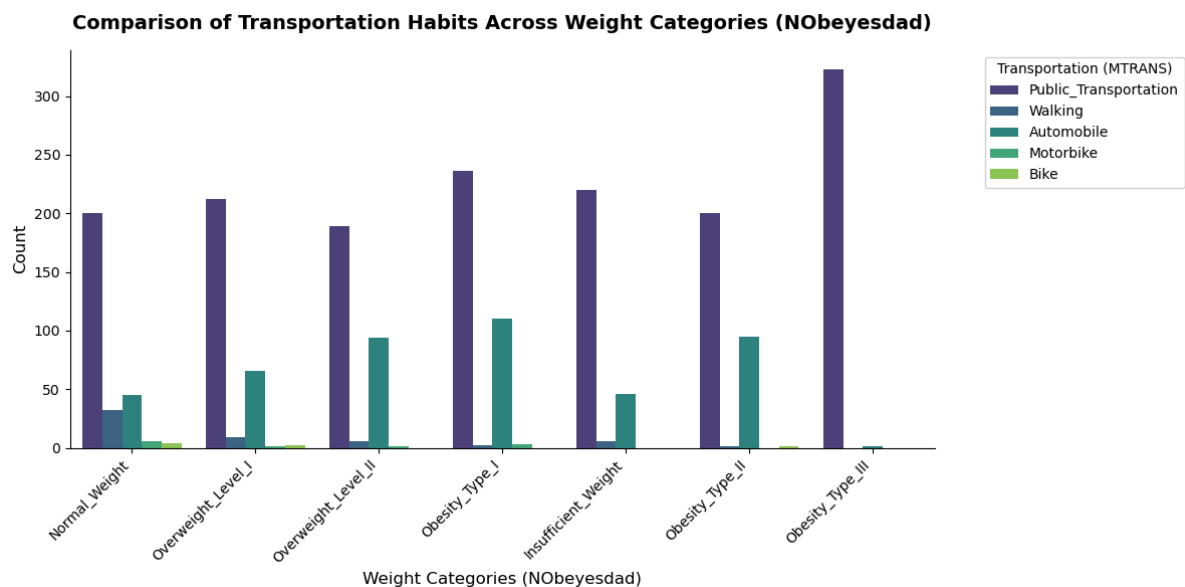
3.3 feature encoding for both binary and multi-class dataset

Categorical encoding is a critical preprocessing step in machine learning because algorithms—particularly tree-based models and regressions—require numerical input to perform mathematical computations and split nodes effectively. Since real-world datasets frequently contain text-based attributes such as gender, transportation methods, or dietary frequencies, leaving these values unencoded prevents models from parsing the data entirely. By transforming categorical variables into structured numerical formats (such as through one-hot or label encoding), I ensured that all relevant behavioral and demographic information was accurately translated into a format the model could evaluate. To perform such a task the pandas' function `get_dummies()` was utilized where: “each variable is converted into as many 0/1 variables as there are different values. Columns in the output are each named after a value; if the input is a DataFrame, the name of the original variable is prepended to the value.”(4)

In particular target variable NObeyesdad was mapped:

- in the binary dataset: 0 (not obese), 1 (obese)
- in the multiclass dataset: going from 0 (insufficient_weight) to 6 (Obesity_Type_III)

Following target verification, the datasets were prepared for machine learning by separating the target variable (NObeyesdad) from the predictor attributes for both the binary and multi-class configurations. Then attention was put into optimizing the feature space. While one-hot encoding with `get_dummies()` was necessary to transform categorical attributes for machine learning compatibility, visual inspections of the unencoded dataset provided vital insight into categorical distributions and variance—such as with transportation modes.



This analysis informed the removal of variables introducing background noise and confirmed that continuous features possessed sufficient variance and discriminative power.

4. Data preprocessing

Preprocessing data often involves filtering out uninformative features to reduce noise and improve model efficiency. By applying a Variance Threshold with a specific cutoff (like 0.05), I was able to automatically drop columns and one-hot encoded features where the vast majority of values are identical, meaning they lacked the variability needed to help my machine learning algorithms learn meaningful patterns. This technique ensured that sparse or near-constant features —such as rarely used transportation categories— were safely removed, resulting in a cleaner, lower-dimensional dataset ready for training.

5. Modeling

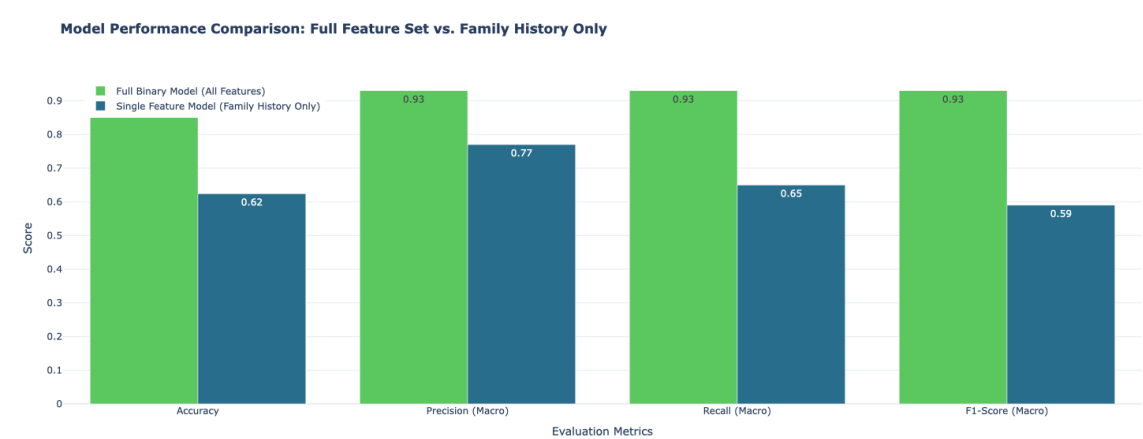
Before training my models i performed `test_train_split()` using the `sklearn.model_selection` module (5) which arbitrary splits the dataset following the 80/20 rule so I configured `test_size=0.2` to ensure my model learns from a robust 80% of the data while saving a dedicated 20% to fairly evaluate its performance on unseen records. To keep my experimentation consistent and reproducible across different runs, I set `random_state=42` so the random shuffling always selects the exact same rows. Finally, I included `stratify=y` to maintain the precise class distribution of my target variable in both subsets, ensuring my model trains and tests on balanced proportions rather than accidentally skewing the evaluation metrics.

I chose to use Random Forest for this dataset because it offers several key advantages that make it particularly well-suited for lifestyle and health-related tabular data. Unlike distance-based algorithms such as Support Vector Machines, Random Forest natively handles a combination of continuous variables and binary encoded attributes without requiring mandatory feature scaling. Furthermore, through its ensemble learning approach (bagging), the algorithm is inherently robust against noise and less prone to overfitting. It excels at capturing complex, non-linear interactions between variables—such as how family history and dietary habits jointly influence health outcomes—while also providing intuitive insights into feature importance.

For the training of the **binary model** i used an approach based on comparison since the research question was:

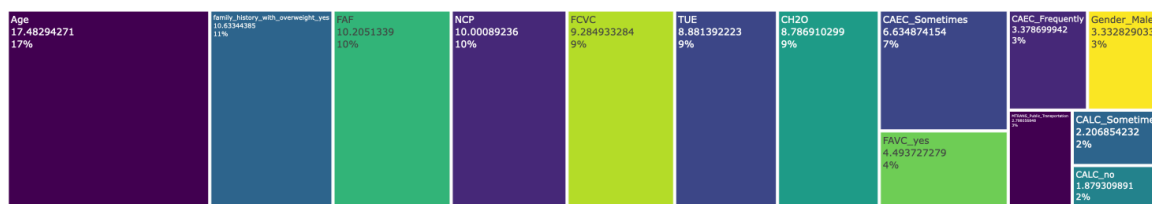
“Can we accurately predict whether someone falls into an 'Obese' category using a binary classification approach, and is family history alone sufficient to make that prediction?”

Therefore, I adopted a comparative approach by evaluating two distinct Random Forest configurations: one trained on the entire feature set and another restricted solely to the `family_history_with_overweight_yes` variable.



The experimental results provide a clear answer: while the full binary model achieves a high accuracy of **0.9338** with balanced macro averages of **0.93** across precision, recall, and F1-score, the single-feature model suffers a sharp drop to an accuracy of **0.6241** and a macro F1-score of **0.59**. This stark performance gap demonstrates that while family history provides a meaningful baseline signal, it is insufficient on its own, proving that a comprehensive set of features is required for accurate and reliable obesity classification.

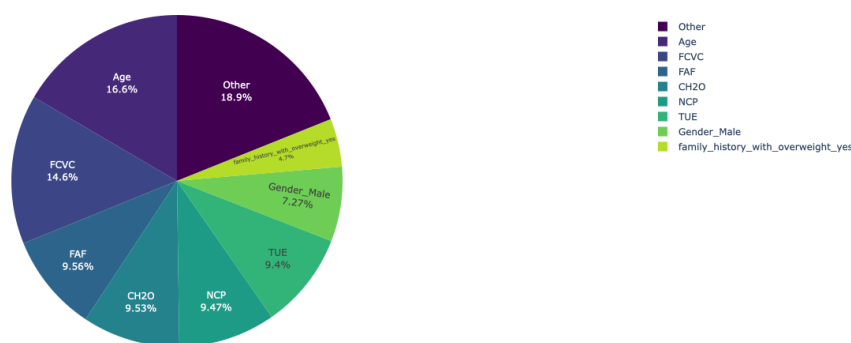
Feature Importances Treemap - Binary Model



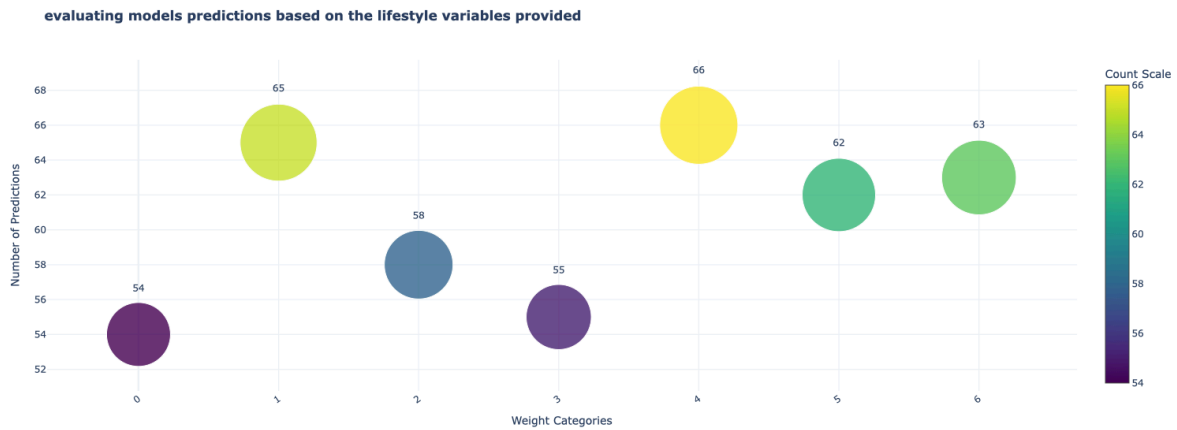
A feature importance treemap was also produced for the binary model evaluation; to better understand what drives the binary model's high accuracy, the feature importances treemap visualizes the relative contribution of each variable. The analysis reveals that **Age** is the single most influential predictor, accounting for **17%** of the model's total importance. Notably, **family_history_with_overweight_yes** emerges as the second most critical feature at **11%**, providing valuable context as to why it served as a meaningful standalone baseline earlier. Other lifestyle and dietary habits—such as physical activity frequency (**FAF** at 10%), number of main meals (**NCP** at 10%), and vegetable consumption (**FCVC** at 9%)—distribute the remaining predictive weight, demonstrating that the model leverages a balanced mix of demographic and behavioral data to classify obesity risk.

For the training of the **multi-class model** as aforementioned the same 80/20 rule on a random Forest classifier was utilized and a visualization of the most important feature was produced

Top Features - Multiclass Model (Feature Importance)



Comparing the feature importance distributions between the binary and the multi-class Random Forest model reveals that while **Age** remains the primary predictor across both frameworks (accounting for roughly 17% in the binary model and 16.6% in the multiclass model), the weight assigned to other features shifts significantly. In the binary model, **family_history_with_overweight_yes** stands out as the second most influential factor at 11%, helping the algorithm draw a broad line between two distinct groups. However, in the multiclass model, its importance drops sharply to 4.7%, while dietary habits like **FCVC** surge to 14.6%. This shift indicates that distinguishing between granular, overlapping obesity categories requires finer behavioral nuances rather than broad risk indicators like family background.



To demonstrate the distribution of model predictions, I produced a bubble chart evaluating model outcomes based on the lifestyle variables provided. This visualization illustrates that predictions are evenly distributed across all seven weight categories—ranging tightly between 54 and 66 counts—confirming that the multiclass model avoids majority-class bias and robustly utilizes lifestyle features across the entire spectrum.

6. Comparing Random Forest with SVM and K-NN

The choice of implying a Random Forest method as a classifier was further validated putting it in comparison with the aforementioned methods being: Support Vector Machines and K-NNs. The comparison was made possible because the initial datasets had been already standardized and scaled on being made only of numerical features, which as we know is a fundamental characteristic of SVM and K-NN train logic. The metrics used to evaluate the performance were found under the `sklearn.metrics` library module of `classification_report` and `accuracy score`.

$$\text{Precision} = \frac{TP}{TP + FP}$$

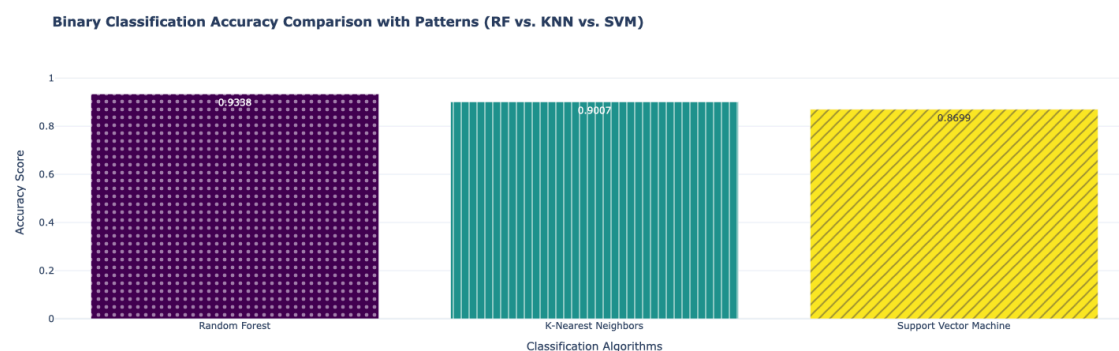
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Accuracy, Precision and Recall are three fundamental metrics used to evaluate the quality of a machine learning model and can aid in the choice of the final selection based on specific trade-offs of each algorithm. In particular, while **Precision** measures the proportion of correctly predicted positive instances out of all instances predicted as positive by the model, **Recall** measures the proportion of correctly predicted positives against all actual positive instances present within the dataset. Finally, **Accuracy** provides an overall measure of correctness by calculating the ratio of total correct predictions to all examined instances.

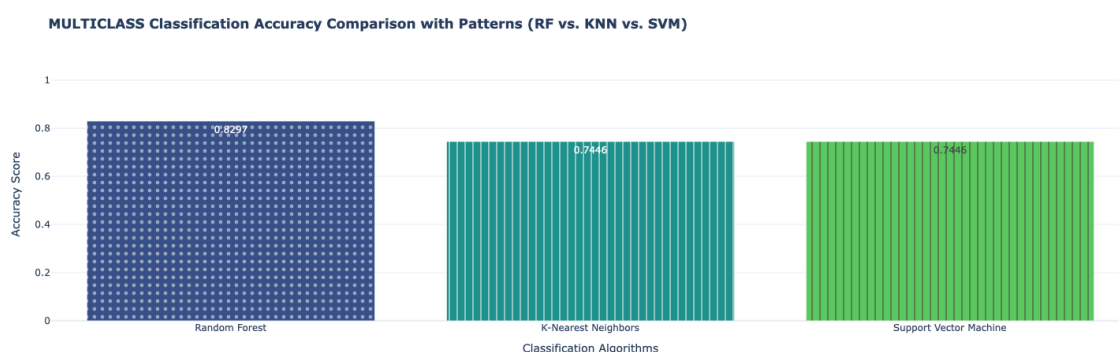
Binary comparison Random Forest-SVM-KNN

Based on the empirical results, Random Forest outperformed both Support Vector Machines (SVM) and K-Nearest Neighbors (K-NN) across all evaluated metrics. Specifically, Random Forest achieved the highest overall accuracy (~93.38%), followed by K-NN (~90.07%) and SVM (~87.00%). This consistent superiority in both Precision and Recall indicates that Random Forest is the most robust and reliable classifier for this dataset, effectively minimizing both false alarms and missed detections across both target classes.



Multi-class comparison Random Forest-SVM-KNN

The same evaluation procedure was used for the multiclass model revealing that Random Forest again outperformed both SVM and K-NN, achieving a significantly higher overall accuracy of ~83.22% compared to the identical 74.46% accuracy shared by both SVM and K-NN. While all three models struggled with Class 1 and performed exceptionally well on Class 6 (with SVM and K-NN matching Random Forest's near-perfect 0.98 F1-score), Random Forest handled the mid-tier classes much more effectively. This pattern demonstrates that Random Forest scales much better to multi-class complexity, maintaining a balanced macro F1-score of 0.83, while SVM (0.74) and K-NN (0.74) experience a more severe drop in predictive performance across the majority of categories.



7. Random Forest model evaluation

The evaluation of the Random Forest model was systematically conducted across both binary and multi-class classification setups to assess its generalization capabilities. The core objective of this dual-evaluation approach was to analyze how effectively the model leverages its trained feature space across varying levels of target granularity. Furthermore, it aimed to determine how robustly the features contribute to the final predictive decisions, ensuring that the model maintains its discriminative power when transitioning from a simple binary choice to a more complex, multi-class environment.

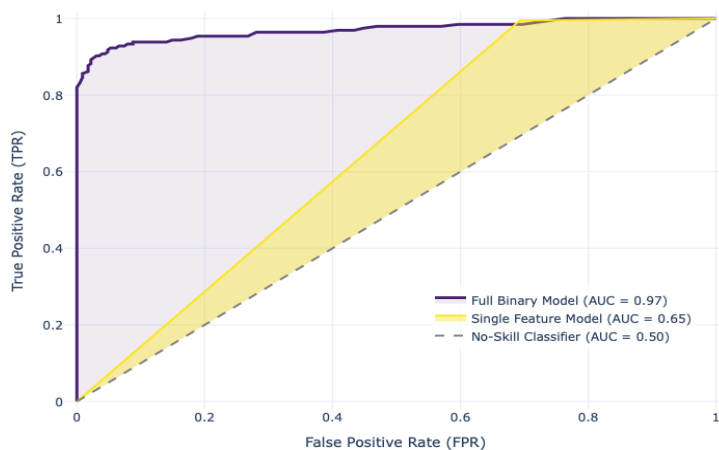
7.1 model evaluation of Binary classifier (full model against one variable only)

Logarithmic loss (log-loss) serves as a rigorous performance metric to evaluate the quality of predicted probabilities rather than discrete classifications. By outputting a continuous score, it measures algorithmic confidence, proportionally penalizing incorrect predictions based on their distance from the true label. While isolating a single predictor—such as `family_history_overweight_yes`—yields a poor baseline

accuracy of approximately 65%, analyzing its information gain within the full feature space provides deeper insights into feature interactions. To quantify this, the log-loss of the comprehensive binary model was compared against a single-feature baseline: this revealed a stark contrast in model certainty. The single-feature model suffers from a high log-loss of 0.5853, underscoring considerable predictive uncertainty when relying solely on family history. Conversely, integrating the full suite of available features dramatically reduces the log-loss to 0.2306. This significant drop demonstrates that the full model successfully leverages collective entropy reduction, resulting in substantially more confident, well-calibrated, and robust probability estimates.

The Receiver Operating Characteristic - Area Under the Curve (ROC AUC) is a standard performance metric used to evaluate a classifier's ability to distinguish between classes across all possible decision thresholds. The ROC curve visually plots the True Positive Rate against the False Positive Rate, illustrating the model's sensitivity versus its false alarm rate. The AUC (Area Under the Curve) collapses this curve into a single scalar value ranging from 0.5 to 1.0, where a score of 0.5 represents a completely random guess and a score of 1.0 indicates a perfect classifier. Because it relies on probability rankings rather than a single fixed classification threshold, ROC AUC provides a robust, threshold-independent assessment of a model's true discriminative power.

Comparison of ROC Curves for Obesity Prediction Models

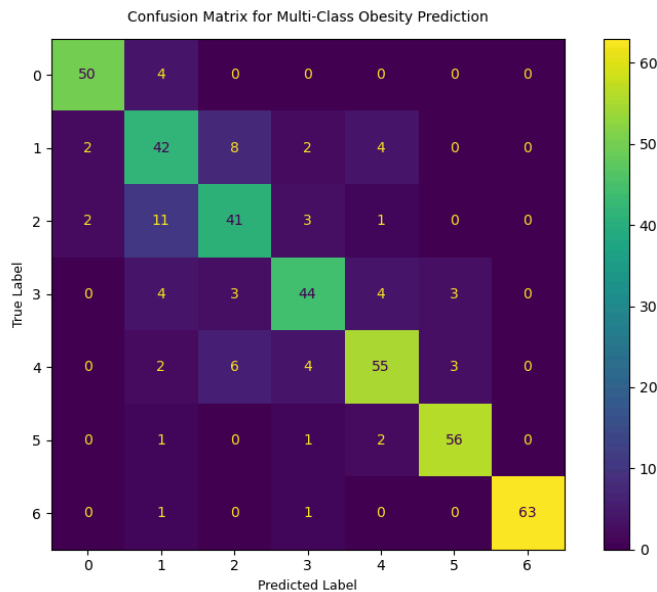


The receiver operating characteristic curve analysis shows that the full binary model achieves an exceptional area under the curve score of 0.97, indicating a near-perfect capability to distinguish between classes. In contrast, the single-feature baseline model yields an area under the curve score of 0.65, performing only moderately better than a random classifier at 0.50.

7.2 model evaluation of Multi-class classifier

Standard ROC curves and AUC metrics cannot be directly applied to multi-class classification because they are fundamentally designed for binary outcomes. The ROC framework relies on plotting a single True Positive Rate against a False Positive Rate, which requires a clear distinction between a solitary "positive" class and a solitary "negative" class. In a multi-class scenario—such as this 7-class obesity prediction task—there is no single opposite class to compute a standard false positive rate against. For this reason, a confusion matrix was implied in order to bypass this binary restriction and directly observe the multi-class performance map.

The Confusion Matrix visually confirms the high accuracy of the model while pinpointing the exact areas of class confusion. Looking at the main diagonal, the model shows an excellent ability to correctly classify Class 6 (63 correct predictions out of 65), Class 5 (56 correct), and Class 4 (55 correct). However, the matrix highlights a significant overlap between Class 1 and Class 2: specifically, 11 instances belonging to Class 2 were misclassified as Class 1, and 8 instances of Class 1 were misclassified as Class 2.



This cross-class confusion explains why Class 1 suffered from the lowest precision and recall scores noted in the report. Overall, the matrix proves that while the model handles extreme categories (like Class 0 and Class 6) almost flawlessly, the mid-tier categories present closer feature boundaries that lead to minor misclassifications.

To further address misclassifications, GridSearchCV was utilized to explore if tuning the *hyperparameters* could enhance the Random Forest's performance. The objective was to systematically test different architectural combinations—varying the number of trees, maximum tree depth, split thresholds, and feature selection subsets—to find the optimal configuration. The search was initialized with a 5-fold cross-validation strategy, optimizing for overall accuracy. However, a comparison of the confusion matrices before and after hyperparameter tuning revealed that the model's predictive behavior remained nearly identical. While tuning led to minor shifts in individual cell counts—such as slight variations in correct classifications for Class 0 and Class 4—the overarching error patterns and diagonal concentrations stayed consistent. This outcome suggests that the baseline Random Forest configuration was already operating near its mathematical ceiling for this specific feature space.

8. Final results and conclusion

The binary classification analysis successfully demonstrated that individuals can be accurately identified within the 'Obese' category utilizing a comprehensive feature set, yielding a **AUC score of 0.97**. This discriminative capability starkly contrasts with the single-feature baseline model relying strictly on family history, which achieved a modest **AUC of 0.65**—performing only marginally better than a random guess. Consequently, while family history serves as a critical genetic and environmental predisposition, it is fundamentally insufficient on its own for reliable classification.

To address our initial question- *Can we accurately predict whether someone falls into an 'Obese' category using a binary classification approach, and is family history alone sufficient to make that prediction?*— we can confidently state that a binary classifier achieves outstanding accuracy when leveraging a multidimensional feature set, but family history on its own completely lacks the necessary discriminative power to make reliable standalone predictions.

Regarding the multi-class question- *Can we accurately predict an individual's obesity level category (NObeyesdad) based on their eating habits and physical condition?* - the analysis investigated whether an individual's specific obesity level (NObeyesdad) could be effectively predicted using behavioral and physiological attributes, such as eating habits and physical condition. The **Random Forest architecture achieved an overall accuracy of 0.83**. An examination of the class-specific metrics and confusion matrices reveals clear structural patterns in how the model operates. Specifically, the model excels at identifying

distinct demographic or physiological extremes—such as *Underweight (Class 0)*, *Obesity Level II (Class 5)*, and *Obesity Level III (Class 6)*—exhibiting near-optimal precision and recall. Conversely, misclassifications are heavily concentrated among adjacent, intermediate categories—such as *true Class 2 (Overweight I) instances occasionally being predicted as Class 1 (Normal weight)*. This pattern reflects the continuous and overlapping nature of clinical obesity grading, where definitions between neighboring weight categories are subtle.

Moreover, a comparative analysis against a hyperparameter-tuned model demonstrated virtually identical predictive distributions and error patterns, with the test accuracy remaining stable at **~0.83**. This stability confirms that the default Random Forest configuration was already robustly optimized to capture the complex, underlying multi-class boundaries without suffering from severe structural bias or overfitting.

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